

BHARAT TECH-SHAKTI MISSION • MASTERCLASS

How Machines Learn, Where They Fail, and What You Can Build

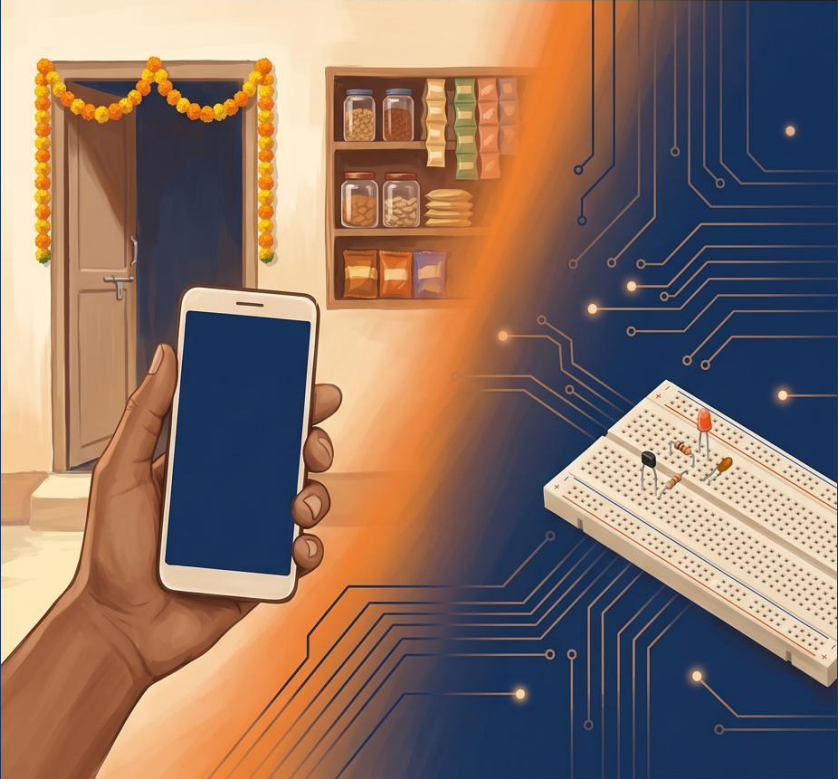
Masterclass on AI

Pranavananda International School, Sector 92, Gurugram

Swami Pranavananda AI & Robotics Lab
Bharat Sevashram Sangha, Srinivaspuri, New Delhi

एआई शक्ति

Everyone is talking about AI. Here you learn to use it,
and build with it, with your own hands.

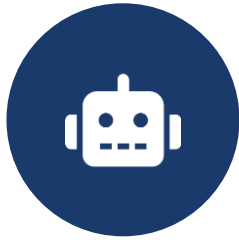


From your first question to your first circuit.
AI and Robotics for every age, starting from zero.

Bharat Tech-Shakti Mission
सृजन से शक्ति

Beginner Course

Hands up if...



**You used an AI tool
this week**



**You used one for
homework**



**You could explain
how it works inside**

Which face is real? Hands up: left or right.



Left



Right

Neither of them exists.

Every hand that went up chose a face that belongs to nobody.

By the end of today you will know how these faces were made, and why that matters for everything you see online.

NEW WORDS

artificial intelligence

generative AI

<https://thispersondoesnotexist.com/>

What we mean by AI

A computer doing a task that we used to think needed human judgement.



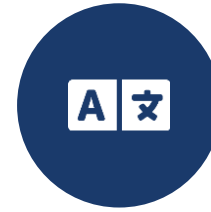
Recognise

a face, a voice, a disease
on an X-ray



Predict

the next word,
tomorrow's rain, a fraud



Translate

Hindi to Tamil, speech to
text



Create

a picture, an essay, a
song, a face

The next two hours

01 Rules or examples? 15 min

04 Machines that see, hear and create 13 min

02 When machines learn the wrong lesson 15 min

05 Vibe coding: apps and devices 32 min

03 Talking machines 18 min

06 Your move 11 min

01

0:10 – 0:25

Rules or examples?

Why you cannot write the rules for a cat, and how a machine learns it anyway.

Write the rules for spotting a cat in a photo

Rule: Four legs

✗ So does a dog

Rule: Whiskers

✗ So does a tiger

Rule: Pointed ears

✗ So does a fox

Rule: Says meow

✗ Photos are silent



**Can anyone write a rule
that never fails?**

You learned from examples, not rules



See many examples

"That is a cat."
"That is a cat too."



Spot a pattern

Shapes and textures
that keep repeating



Make a guess

"Cat!"
(It was a dog.)



Get corrected

Adjust, and guess
better next time

Repeat thousands of times. That is machine learning.

NEW WORDS

machine learning

You already do this before every board exam



Training data

The past papers
you practised



Algorithm

The method you
used to study



Model

Your prepared
mind on exam
morning



New data

This year's paper,
which you have
never seen

NEW WORDS

training data

algorithm

model

Live: teach a machine in two minutes

- 1 Make three classes: thumbs up, open palm, nothing
- 2 Two volunteers hold each gesture to the webcam
- 3 Record about 30 pictures per class
- 4 Press Train. Wait about 20 seconds
- 5 **Test it. Then try to fool it**



NEW WORDS

pattern recognition

0:25 – 0:40

02

When machines learn the wrong lesson

A model learns whatever is in its examples, including the unfairness hidden in them.

Quick. Five seconds. Write on paper.



1. A name for a nurse



2. A name for a doctor

Do not think about it. Just write the first name that comes.

Where did those names come from?

Nobody taught you a rule.

You picked it up from hospitals you visited, films, advertisements, textbooks and news.

That was your training data.

A 2016 study of word associations learned from Google News articles

**Man is to computer programmer
as woman is to...**

homemaker

Bolukbasi and others, NeurIPS 2016

A model reads the same world you do, only far more of it.

NEW WORDS

bias

Live: the translator must choose he or she

वह डॉक्टर है।  ____ is a doctor.

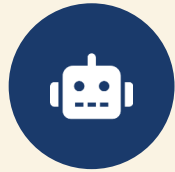
वह नर्स है।  ____ is a nurse.

Which word will the machine choose for each?

The hiring tool that learned an old habit

10

years of past CVs used as
training data, most of them
from men



Amazon built a tool to score job applicants from one to five stars.



It marked down CVs containing the word "women's", as in "women's chess club captain".



Engineers could not prove it was fair. The project was dropped.

Reuters, October 2018

Data is a record of the past.

A model repeats the past unless someone checks it.

Two minutes with the person next to you



An AI will recommend each Class 10 student's stream. It is trained on ten years of this school's records of who did well in which stream. What could go wrong?

Think about who was steered where, what marks leave out, and who is missing from the records.

What this exercise proves



Old habits repeat

If girls were steered toward humanities in the past, the model steers them there too.



Marks miss things

It cannot see who was happy in a stream, or who grew later.



The missing stay missing

A student who never got to try science leaves no record to learn from.



It proves itself right

Students pushed into a stream do well or badly partly because they were pushed.

Data records the decisions people made, not the right ones.

03

0:40 – 0:58

Talking machines

ChatGPT, Gemini and Claude are very good at one thing: guessing the next word.

Your phone keyboard already does this



Your keyboard guesses the next word from what people usually type.

A large language model does the same thing, one piece at a time, after reading a vast library of books, websites and code.

NEW WORDS

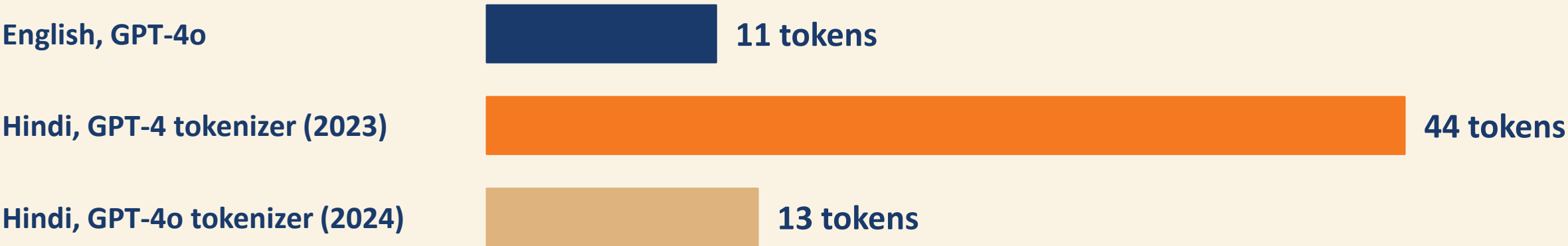
large language model

Models read in pieces called tokens



Each number is that piece's ID in the model's vocabulary. Split by the GPT-4o tokenizer. Other models cut text differently.

The same sentence in Hindi: प्रणवानंद इंटरनेशनल स्कूल गुरुग्राम में है



NEW WORDS

token

Six court cases. None of them existed.

6 → 0

cases cited by ChatGPT, cases
that were real



New York, 2023. A lawyer used ChatGPT to find earlier cases for his client's lawsuit.



He asked ChatGPT if the cases were real. It said yes.



The judge fined the lawyers and their firm 5,000 US dollars.

Mata v. Avianca, New York, June 2023. In India: a Bengaluru tax tribunal recalled an order in January 2025 after it cited four judgments that did not exist.

NEW WORDS

hallucination

It predicts. It does not look up.



A search engine

Finds pages that already exist and shows you where they came from.



A language model

Writes words that are likely to come next. Likely is not the same as true.

If it can invent a court case, it can invent a source for your school project.

The four parts of a good prompt

1

Who you are

I am a Class 11 student who has not studied biology since Class 10.

2

What you want

Explain how vaccines train the body.

3

What shape

Five sentences. Use one everyday comparison.

4

How to check

Then ask me two questions to test whether I understood.

Weak: "Explain vaccines."

Strong: all four parts together

NEW WORDS

prompt

Live: same question, two prompts

Prompt A

"Explain [topic]."

Watch for: length, jargon, whether it fits you.

Prompt B

*"I am a Class 11 [stream] student.
Explain [topic] in five sentences with
one everyday comparison. Then ask
me two questions to check I
understood."*

Topic chosen by the room, from any subject.

0:58 – 1:11

04

Machines that see, hear and create

How a computer reads a photo, understands a voice, and makes a face that never existed.

To a computer, a photo is only numbers

255	255	255	255	255	255	255	255
255	255	0	255	255	0	255	255
255	255	0	255	255	0	255	255
255	255	255	255	255	255	255	255
0	255	255	255	255	255	255	0
255	0	255	255	255	255	0	255
255	255	0	0	0	0	255	255
255	255	255	255	255	255	255	255

0 = black 255 = white

The machine finds patterns in this grid of numbers, just as our Teachable Machine did with your hands.



Face unlock on your phone



DigiYatra face check at Indian airports



Traffic cameras reading number plates

NEW WORDS

computer vision

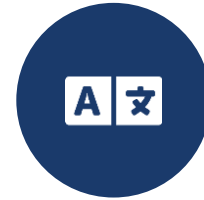
Live: speak one Indian language, hear another



You speak
in Hindi



Speech to text
the words are
written out



Translation
into Tamil, Bengali
or another



Text to speech
read aloud in the
new language

Bhashini: the Government of India's language platform, from MeitY

NEW WORDS

speech recognition

Back to the faces that belong to nobody



The machine studied tens of thousands of photos of real faces. It never copied one. It learned what faces look like, and then made a new one.



Text

essays, code, poems



Pictures

from a one-line
description



Voices

copied from a short
recording

When a face is used without permission

Oct 2023



A video is posted on an Instagram fan page. An actor's face has been put on another woman's body.

6 Nov 2023



It goes viral. The actor Rashmika Mandanna says publicly how frightening it is.

7 Nov 2023



MeitY reminds social media platforms: remove such content within 24 hours of a complaint.

Jan 2024



Delhi Police arrest a 23-year-old engineering student. Police say he made it to gain followers.

NEW WORDS

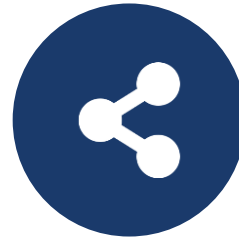
deepfake

Three rules for deepfakes



Do not make one

of a real person without their permission. Not even as a joke.



Do not forward one

Every share spreads the harm further.



Report it

cybercrime.gov.in or helpline 1930. Tell an adult you trust.

1:11 – 1:43

05

Vibe coding: apps and devices

Describe what you want in plain words. The AI writes the code. Your name goes on the result.

Describe it in plain words. AI writes the code.



You describe

"Make me a Tic-Tac-Toe game with my name on it."



AI writes

A complete program, in under a minute



You run it

On a screen or on a small board. Test it. Ask for changes.

The name comes from Andrej Karpathy, February 2025

NEW WORDS

vibe coding

Live build: a game with a student's name on it

ROUND 1: THE FIRST PROMPT

You are a web developer. Build a two-player Tic-Tac-Toe game for me, **[Your name]**. Put "**[Your name]**'s Tic-Tac-Toe" as the page title and the heading. It must work on a mobile phone. Build this as a single HTML file.



Needed

One student lends their name. Two more come up to play on the big screen.

Iterative prompting: one change at a time

Round 2

"Add a score counter for X and O, and a Play Again button."

Round 3

"Use navy and orange colours, and highlight the winning line."

Round 4

*"Add a footer: Made by **[Your name]**, Pranavananda International School."*

Describe

Generate

Run

Test

Iterate



Small, specific changes. Test after every round.

NEW WORDS

iterative prompting

Tonight: a quiz with your name on it

PHOTOGRAPH THIS PROMPT

You are a web developer. Build a 5-question quiz called "[Your name]'s AI Quiz" on these words: bias, token, hallucination, deepfake, vibe coding. Show one question at a time with four options and a green tick for a correct answer. End with "You scored X out of 5 on [Your name]'s AI Quiz". It must work on a mobile phone. Build this as a single HTML file.



Make it yours

Swap the five words for any chapter you are studying. Or make it "How well do you know me?" and send it to your friends.

Get it onto your phone, and keep it



Preview it

Most AI chat apps can show the page running, right inside the chat.



Save it

Copy the code into a notes app. Save it as yourname.html. Open it in a browser.

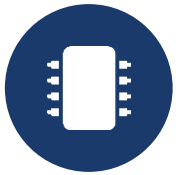


Share it

Send the .html file on WhatsApp as a document. It opens in any browser.

One file. No app store, no installation, no cost.

Meet the board: a tiny computer with Wi-Fi



NodeMCU (ESP8266)

A microcontroller with Wi-Fi built in. It runs one program, in a loop, for as long as it has power.



0.96-inch OLED screen

128 by 64 pixels. Four wires connect it to the board.



Power bank

Once the program is on the board, no laptop is needed.



Pass it around

The newer ESP32 version: two processor cores, more memory, and the screen built in with no wiring.

NEW WORDS

microcontroller

Device build 1: an internet clock in five minutes

THE PROMPT

Write an Arduino sketch for a NodeMCU ESP8266 with a 0.96-inch SSD1306 OLED on I2C (SDA on D2, SCL on D1, address 0x3C). Connect to Wi-Fi, get the time from `in.pool.ntp.org`, and show Indian Standard Time and today's date in large text. Update every second.

- 1 Paste the prompt into the AI
- 2 Copy the code into Arduino IDE
- 3 Type the Wi-Fi name and password
- 4 Press Upload
- 5 Watch the clock appear

Device build 2: planes over this school, live



The aircraft

Broadcasts its position, height and flight code by radio, every second



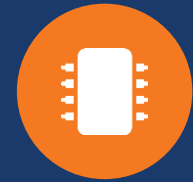
Ground receivers

Volunteers' antennas pick up the signal and upload it



Open database

adsb.fi shares the combined data, free, with no account



Our board

Asks every 10 seconds for planes near the school and shows them

IndiGo

6E 2456 A321neo
3,200 ft flying N

Illustrative screen. This build was done in advance. It took an evening of back-and-forth with the AI, not five minutes.

NEW WORDS

sensor

IoT

What went wrong along the way



The data site said no

The first flight-data site returned error 403: access refused. The AI suggested a different free source.



Two missing characters

A copy-paste dropped the `/*` on line 1. The compiler then misread the whole file and threw a wall of errors.



A wrong first guess

The AI first suspected a clash between code libraries. Printing more detail found the real cause.

Vibe coding is a conversation, not a single command. You still own what it builds.

Three questions before you share it



Can you understand it?

Ask the AI to explain every part until you could fix it yourself.



Can you explain it?

Every choice was yours: the idea, the features, the colours. Own them.



Can you disclose it?

Say it plainly: "AI wrote the code. I directed what it should do."

06

1:43 – 1:54

Your move

What you now know, where it leads in every stream,
and one thing to build this month.

Words you now own

artificial intelligence

generative AI

machine learning

training data

algorithm

model

pattern recognition

bias

large language model

token

hallucination

prompt

computer vision

speech recognition

deepfake

sensor

microcontroller

IoT

vibe coding

iterative prompting

20 words. Each one has a story behind it.

AI is arriving in every stream



Science

AlphaFold predicted the shapes of proteins, a problem biologists had worked on for fifty years. Its makers shared the 2024 Nobel Prize in Chemistry.



Commerce

Banks and payment apps use models that notice unusual spending and stop a payment for a check before the money goes.



Humanities

Translation across Indian languages, and the hard questions: who is responsible when AI is wrong? Law, ethics and journalism need answers.

Your 30-day challenge: build one small thing



Teach a machine

Train Teachable Machine to sort something at home: fruits, coins, your pet's moods.



Vibe-code a device or an app

A desk clock on a small board. A quiz for a younger sibling. A study-hours tracker.



Teach a grandparent

Show them how to photograph a letter and have AI explain it in their language.

**Then show it to
your family.**

Explaining what you built is
how you find out whether
you understood it.

Keep learning

teachablemachine.withgoogle.com

Train your own models, free, in a browser

futureskillsprime.in

FutureSkills PRIME: MeitY's platform for courses in AI and emerging technologies

bharattechshakti.org

The AI Shakti Beginner Course and the Srinivasapuri lab



**Coming soon: AI community
learning at Bharat Sevashram
Sangha, Gurugram**

Help teach it. Volunteers welcome,
with a parent's permission.

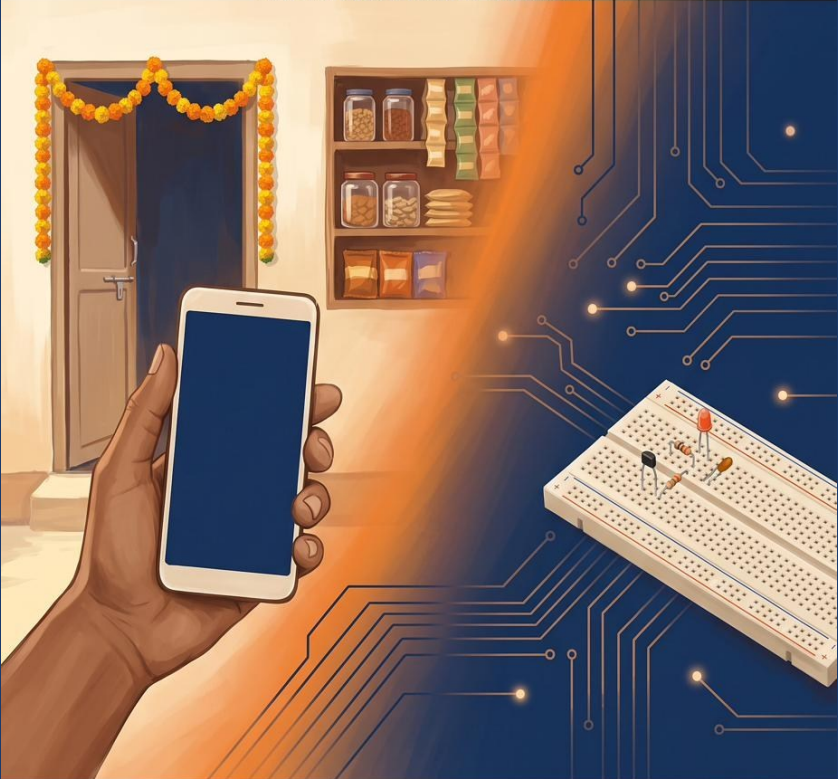
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Questions?

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